

surfacehop_jax: Differentiable, GPU-parallel fewest-switches surface hopping in JAX

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Summary

`surfacehop_jax` is a Python package for nonadiabatic molecular dynamics that combines Tully’s fewest-switches surface hopping (FSSH) algorithm (Tully 1990) with the JAX automatic-differentiation framework. The entire propagator — velocity-Verlet for nuclei, exact matrix exponential for the electronic TDSE, the fewest-switches hop decision, momentum rescaling, frustrated-hop reflection, and the Granucci-Persico energy-based decoherence correction — is implemented as one pure, ~ 400 -line JAX program. Three properties follow at no extra implementation cost: (i) the full integrator compiles to a single XLA kernel via `jax.jit`; (ii) an ensemble of N Wigner-sampled trajectories runs as a single `jax.vmap`, scaling transparently from laptop CPUs to GPUs without code changes; and (iii) reverse-mode autodiff flows end-to-end through the propagator, making gradients of ensemble observables with respect to Hamiltonian parameters available at one-backward-pass cost.

The package ships a `LinearVibronicCoupling` class that constructs the standard Köppel-Domcke-Cederbaum diabatic Hamiltonian (Köppel, Domcke, and Cederbaum 1984) from user-supplied vertical excitation energies, normal-mode frequencies, intrastate gradients (κ), and interstate couplings (λ), together with a factory function returning the canonical four-mode pyrazine S_1/S_2 benchmark used throughout the nonadiabatic-dynamics literature (Worth, Meyer, and Cederbaum 1998). A companion documentation set — an eleven-chapter user guide and a worked Jupyter notebook for gradient-based parameter fitting — accompanies the source.

FSSH is the workhorse trajectory-based method for simulating coupled electronic-nuclear dynamics in photochemistry: ultrafast internal conversion, radiationless relaxation through conical intersections, photoisomerisation, and excited-state charge transfer. Established FSSH codes such as SHARC (Mai, Marquetand, and González 2018), NEWTON-X (Barbatti et al. 2007), JADE (Du and Lan 2015), and PYXAID (Akimov and Prezhdo 2013) have driven a generation of computational photochemistry, but they predate the modern automatic-differentiation and accelerator stack: they are written in Fortran or coupled to ab-initio back-ends through thousand-line per-trajectory drivers and are not designed to be embedded inside JAX-style learning pipelines. `surfacehop_jax` is positioned to complement those codes for the class of problems — model Hamiltonians, parameter fits, ML-trained potentials — where differentiability and GPU parallelism are the dominant figures of merit.

Statement of need

The intersection of machine learning and excited-state chemistry is expanding rapidly: ML-trained potentials, ML-predicted vibronic couplings, and differentiable dynamics workflows

are all active research directions. The trajectory-based dynamics codes the community relies on were not designed for this setting. Three concrete pain points motivate `surfacehop_jax`:

1. **GPU-parallel ensembles.** A converged FSSH calculation typically needs 10^3 - 10^4 independent trajectories. Existing codes parallelise by spawning subprocesses, each with its own input file and Python/Fortran-driven event loop. `surfacehop_jax` runs the entire batch as one `jax.vmap`-batched XLA kernel; wall-clock cost scales as one trajectory plus a small constant, not as N trajectories. On a single consumer GPU this puts 10k-trajectory ensembles within reach for ordinary photochemistry problems.
2. **Differentiable dynamics.** Inverse problems in nonadiabatic dynamics — fitting LVC parameters against transient-absorption signals, training ML potentials whose downstream FSSH observables match a quantum reference, sensitivity analysis on model couplings — all require gradients of final-state observables with respect to Hamiltonian parameters. These gradients are not available in existing FSSH codes; reproducing them by hand requires deriving and coding the full adjoint of the propagator. `surfacehop_jax` inherits them for free from JAX’s reverse-mode autodiff at $O(1)$ cost in the number of parameters.
3. **Built-in vibronic-coupling convenience.** Setting up an LVC Hamiltonian from scratch is currently tedious: one writes a custom diabatic-matrix function, computes its gradient analytically or numerically, and threads it into the dynamics driver. With `LinearVibronicCoupling` the user supplies frequencies and coupling tensors as plain arrays and the package produces a JAX-traceable Hamiltonian whose adiabatic energies, gradients, and non-adiabatic couplings are computed internally via autodiff. Combined with the companion package `nma_jax` for normal-mode analysis from quantum-chemistry Hessians, this gives a fully Python end-to-end workflow from ab-initio gradients to ensemble surface-hopping dynamics.

Implementation

The diabatic-to-adiabatic transform is one autodiff call: given any function $H(\mathbf{x})$ returning a Hermitian $(n_{\text{el}}, n_{\text{el}})$ matrix, `jax.jacrev(H)(x)` produces the gradient tensor $\partial H / \partial x$, and a single einsum with the eigenvectors of $H(\mathbf{x})$ yields adiabatic energies, Hellmann-Feynman gradients, and non-adiabatic coupling vectors in one pass. Eigenvector phases are tracked between steps by sign-aligning with the previous step’s eigenvectors, preserving the continuity of the coupling tensor along a trajectory. The electronic equation of motion is integrated by exact matrix exponentiation (`jax.scipy.linalg.expm`) over each nuclear step, which gives an unconditionally stable propagator for the constant-generator limit and side-steps the still-poor complex-valued ODE support in SciPy.

The fewest-switches hop probability is computed in the standard form (Tully 1990); for $n_{\text{el}} \geq 3$ the unbiased cumulative-probability selection scheme is used. Momentum rescaling on a successful hop is along the non-adiabatic coupling direction with the Hammes-Schiffer / Tully root selection (Hammes-Schiffer and Tully 1994); frustrated hops trigger Truhlar’s velocity-reversal prescription (Jasper and Truhlar 2003). The Granucci-Persico EDC (Granucci and Persico 2007) in the Zhu-Truhlar form (Zhu et al. 2004) is exposed via a `decoherence_fn` keyword on `step`, `simulate`, and `run_ensemble`, leaving the bare-FSSH path bit-identical to a no-correction baseline.

The propagator is a pure function of a `TrajectoryState` named tuple, so the same code drives 1D Tully models, multi-mode LVC Hamiltonians, and any future ab-initio coupled back-end. Ensembles are `jax.vmap(simulate)`. The complete source — propagator, model classes, Wigner sampler, decoherence corrections, and tests — totals approximately 1000 lines.

Validation

Three benchmarks ship with the package and run in CI:

1. **Tully 1990 Model 1.** The single-avoided-crossing transmission curve is reproduced over 25 momenta with 500 trajectories each, in agreement with Tully’s original published values (Tully 1990) to within statistical noise. FSSH internal consistency (trajectory fraction $\approx \langle |c_2|^2 \rangle$) is preserved across the scan, confirming the sign convention in the hop probability and the correctness of the momentum-rescaling and frustrated-hop branches.
2. **Pyrazine $S_2 \rightarrow S_1$ ultrafast internal conversion.** The four-mode Köppel-Domcke-Cederbaum model is propagated for 120 fs starting from a Wigner-sampled Franck-Condon distribution on the bright S_2 state. Bare FSSH reproduces the canonical ~ 20 fs initial decay and exhibits the well-known over-coherence artefact (a gap between the active-state fraction and $\langle |c|^2 \rangle$). With EDC enabled the two diagnostics overlap (internal consistency restored) and the S_2 decay timescale shifts to ~ 50 fs, in better agreement with high-level quantum reference dynamics on the same model.
3. **End-to-end differentiability.** A companion notebook (`notebooks/differentiable_dynamics.ipynb`) verifies that `jax.grad` correctly differentiates a Wigner-sampled FSSH ensemble with EDC through `jax.lax.scan` over time and `jax.vmap` over the ensemble. The gradient of an analytic eigenvalue derivative agrees with `jax.grad` to $\sim 7 \times 10^{-18}$ (the float-64 round-off floor). On a 30-trajectory pyrazine ensemble propagated for 12 fs under EDC, the autodiff gradient of $\langle |c_{S_2}|^2 \rangle$ with respect to the interstate coupling λ_{10a} agrees with centred finite-difference to a relative error of 3×10^{-5} . An Adam-based optimisation of λ_{10a} against a synthetic target observable converges in 20 iterations, reducing the loss by approximately five orders of magnitude. A simultaneous five-parameter gradient (one excitation energy, three intrastate gradients, one interstate coupling) is obtained in a single backward pass, demonstrating the $O(1)$ -in-parameter-count scaling that makes high-dimensional LVC fitting tractable.

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